

Skills Summary

Area Models for Two-Digit Multiplication

Use this reference while completing your worksheet or when studying.

Skill 1 — Build the Model (Split the Factors)

Rule: split each factor into tens + ones and put one split along each side of the rectangle. The splits must add back to the original factors.

Example: Split 28 for an area model: $28 = 20 + ?$

Step 1. A two-digit number splits into its tens and its ones; 28 has 2 tens already written as 20.

Step 2. $28 - 20 = 8$, so the ones part is 8.

$\Rightarrow$ 8

Try it: $63 = 60 + ?$

check: 8

Skill 2 — Find Each Partial Product

Rule: each cell multiplies one part of each factor — tens $\times$ tens, tens $\times$ ones, ones $\times$ tens, ones $\times$ ones. Use the basic fact, then attach a zero for every ten.

Example: In 34×12 , what is the largest cell?

Step 1. The largest cell multiplies the two tens parts: 30 and 10.

Step 2. $30 \times 10 = 300$ (fact 3×1 , plus two zeros).

$\Rightarrow$ 300

Try it: In 26×13 , what is the largest cell?

check: 200

Skill 3 — Add the Cells to Get the Total

Rule: the product is the **sum of all four cells** — the whole rectangle. Leaving out a cell is the classic area-model mistake.

Example: Finish 21×34 from its model.

Step 1. Cells: $20 \times 30 = 600$, $20 \times 4 = 80$, $1 \times 30 = 30$, $1 \times 4 = 4$ — all four must be added.

Step 2. $600 + 80 + 30 + 4 = 714$

$\Rightarrow$ 714

Try it: Use a model to find 12×13 .

check: 156

Skill 4 — Estimate to Check

Rule: round each factor to the nearest ten and multiply the rounded numbers. If the exact answer is far from the estimate, hunt for a missing cell.

Example: Estimate 29×41 .

Step 1. Round each factor to the nearest ten: 29 goes to 30 and 41 goes to 40.

Step 2. $30 \times 40 = 1,200$

$\Rightarrow$ **1,200**

Try it: Estimate 52×38 the same way.

check: 2,000